EE 354 – SYNCRONOUS GENERATOR

PERERA J.D.T.

E/21/291

GROUP EE.21.15

SEMESTER 05

13/01/2026

EE-354 SYNCRONOUS GENERATOR

E/21/291

1. What is an electrical Machine?

Electrical machines can be defined as devices that converts electrical energy in to mechanical energy or vice versa. most commonly motors and generators are examples of electrical machines. Moreover electrical machines are of 2 types AC and DC machines.

2. What are the 2 main components of a synconous generator?

Stator - stationary part consist of armature winding
Rotor - Rotating Part generates the magnetic field.

3. What is the main difference between syncronous machine and induction machine?

Syncronous machines operate at syncronous speed, which means that the rotor speed is equal to the speed of rotating magnetic field.

Induction motors and generators operate on electromagnetic induction. The rotor does not get power directly. It rotates due to the induced current caused by the stators rotating magnetic field.

4. What are the type of rotors used in synconous generators?

Cylindrical rotor - Cylindrical steel shaft with embeded windings. used for high speed generators and motors.

Salient pole rotor - poles of the generator are projected outwards. used in low speed generators such as in hydro power stations.

Scanned with CamScanner

What are the methods used to supply DC voltage to the rotor winding?

Slip rings and brushes - DC current flows to the rotor field winding through carbon brushes that makes contact with slip rings.

Brushless and exitation - A small exitation generator is mounted on the same shaft as the main generator.

6. what is the drawback of slip rings and brushes method?

The carbon brushes wear down over time and needed to be replaced. Sparking can occur at high speeds and mechanical contact creates heat.

7. How is voltge generated in stator windings when the load is zero?

When the rotor is exited with DC current and rotated at synchronous speed the magnetic flux created by it cuts the stator windings. Therefore according to faraday's law The changing magnetic field induces a voltage in the stator.

8. what are the armature reactions?

Armature reaction is the effect of magnetic field produce by the stator on the main magnetic field produced by the ~~stator~~ rotor. This tends to magnetize and demagnetize the main field.

9. what are the types of losses in a synchronous generator?

copper losses - heat generated in the stator and rotor
core losses - hystresis and eddy current
mechanical losses - friction in bearings
Stray losses -

Scanned with CamScanner

what are the tests carried out to find parameters of a synchronous generator?

Open circuit test
Short circuit test
DC test

11. what data are available on the nameplate of a synchronous generator?

Rated voltge
Rated current
Rated power
Power factor
frequancy
speed
manufacturer details

12. what are the advantages of operating synchronous generators in parallel?

improved reliability
load sharing
Ease of maintanance
Improved operating efficiency.

13. what methods are used to identify whether phase sequence is the same?

Phase sequance indicator
Three lamp method
Two lamp method

14. what is a synchronoscope?

A synchronoscope is an instrument used to indicate whether the incoming generator is synchronized with the bus bar by checking phase sequence phase angle difference.

Scanned with CamScanner

EE 354 POWER ENGINEERING
SYNCHRONOUS GENERATOR
Instruction Sheet

AIM

1. To calculate leakage reactance, armature reactance and synchronous reactance using open circuit and short circuit characteristic curves.
2. To calculate the short circuit ratio of the generator.
3. To draw the performance chart of a synchronous generator

APPARATUS

1. Synchronous Generator (SM 100-3, 1425 rpm, Armature-230V ac/1.6A, Field-220V dc/0.6A)
2. DC Shunt Motor (DM-100A, 1500 rpm, Armmature-220V dc/1.4A, Shunt Field- 0.5A)
3. AC Voltmeter (0-750 V)
4. AC Ammeter (0-2 A)
5. DC Ammeter (0-1 A)
6. Inductive loads (4.56 kVzA, 50 Hz, 230 V, 19 A)
7. Digital Tachometer

DATA

For Synchronous Generator

1. Rated Voltage - 400V Line
2. Rated Current – 1.6 A Line
3. Maximum Safe Field Current – 0.6 A

PROCEDURE

1.Connect the circuit as shown in Figure 1.
2.Get the circuit checked by the instructor before continuing the practical.

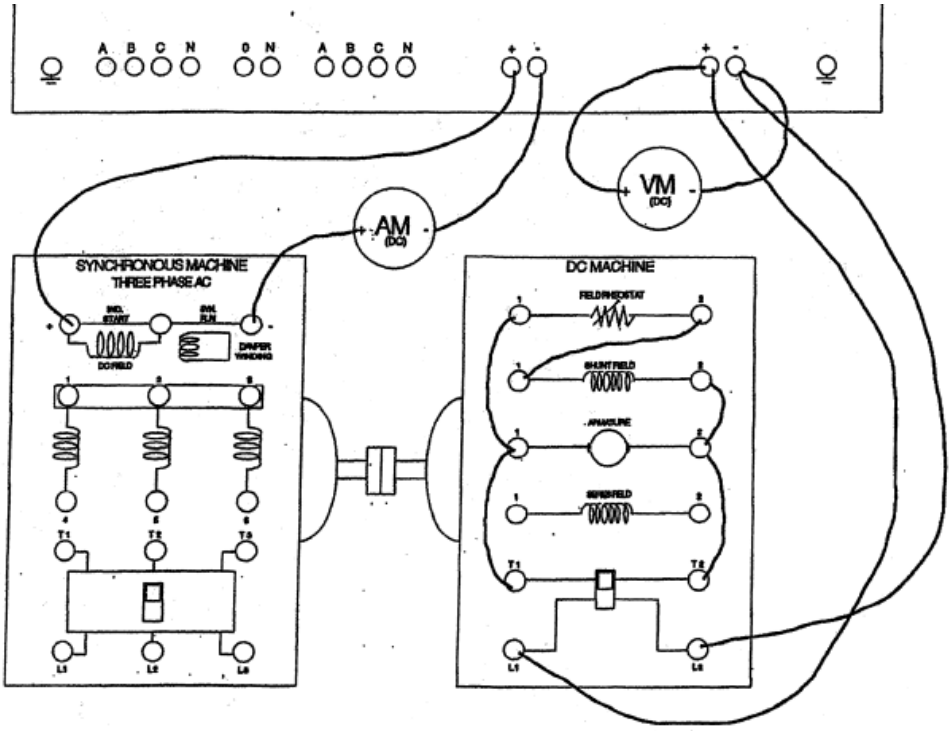


Figure 1: Connection diagram of synchronous generator and DC motor

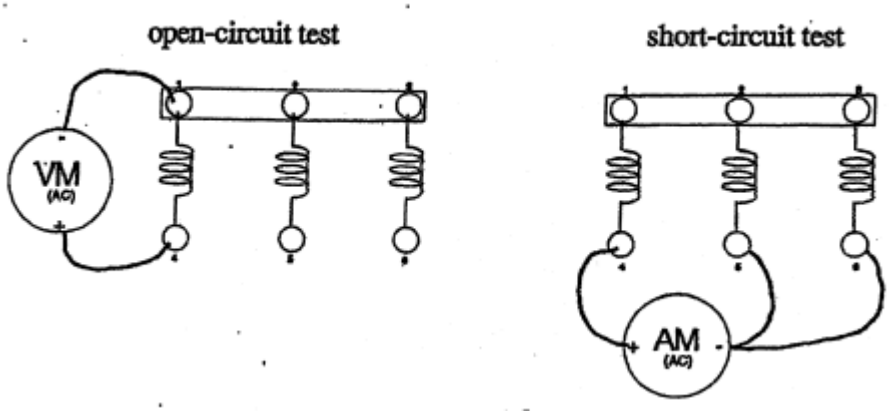


Figure 2: Synchronous generator terminal connections for open circuit test and short circuit test

Part 1: Open Circuit Test

1. Connect the generator terminals as in Figure 2 for open circuit test.
2. Get the circuit checked by the instructor.
3. Set the 220V DC knob to zero.
4. Set the 0-150V DC knob to zero.
5. Set the synchronous machine to 'IND START'.
6. Bring the DC machine terminal voltage up to 220V and adjust the field rheostat to obtain a speed of 1500 rpm.
7. Set the synchronous machine to 'SYNCH RUN'.
8. With the synchronous machine's terminals open, measure the voltage across the windings (Voc) as field current (If) is increased from 0 to 0.6A in steps of 0.1A. (The field current is increased by increasing the DC voltage across the field winding.)
   Special care must be taken not to exceed excitation current more than 0.6A
9. Adjust the excitation current to **zero**.

10. Make sure the generator terminal voltage is **zero** by setting synchronous machine to 'IND START'.
11. Set the 220V DC knob to zero.

Part 2: Short Circuit Test

1. Short circuit the generator terminals and connect an ammeter in series with the shorted windings as shown in Figure 2.
2. Get the circuit checked by the instructor.
3. Set the 220V DC knob to zero.
4. Set the 0-150V DC knob to zero.
5. Set the synchronous machine to 'IND START'.
6. Bring the DC machine terminal voltage up to 220V and adjust the field rheostat to obtain a speed of 1500 rpm.
7. Set the synchronous machine to 'SYNCH RUN'.
8. Take the average winding short circuit currents (Isc) as excitation current (If) increased from 0 to 0.6A in steps of 0.1A. (The field current is increased by increasing the DC voltage across the field winding.)

   Special care must be taken not to exceed the generator rated current. (1.6 A)
9. Adjust the excitation current to **zero**.
10. Make sure the generator terminal voltage is **zero** by setting synchronous machine to 'IND START'.
11. Set the 220V DC knob to zero.

OBSERVATIONS

Table1: Open Circuit Test

Field Current, I_f/A	Open Circuit Voltage, V_{OC}/V
0	
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	

Table 2: Short Circuit Test

Field Current, I_f/A	Short Circuit Current, I_{SC}/A
0	
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	

 Plot the open circuit and short circuit curves in the same graph.

CALCULATIONS

1) Assuming high X/R ratio, calculate synchronous reactance (saturated & unsaturated) using open circuit and short circuit curves.

   Refer: George McPherson, "Electrical Machines and Transformers", 2nd edition, John Wiley & Sons, Pages 107- 121
2) Find the value of Short Circuit Ratio (SCR) of the machine.
3) Find the voltage regulation at full load at unity power factor and 0.8 power factor (lagging) using a) Synchronous impedance method b) M.M.F method

4) Draw the performance chart of the synchronous generator.

DISCUSSION

1. What are the factors to be considered when connecting a synchronous generator to the national grid?
2. Mention methods to verify the phase sequences of the generator and national grid?
3. Discuss how can you change frequency of the generated voltage?

Refer: S.J. Chapman, "Electrical Machinery Fundamentals", Second Edition, McGraw-Hill, Pages 149-152

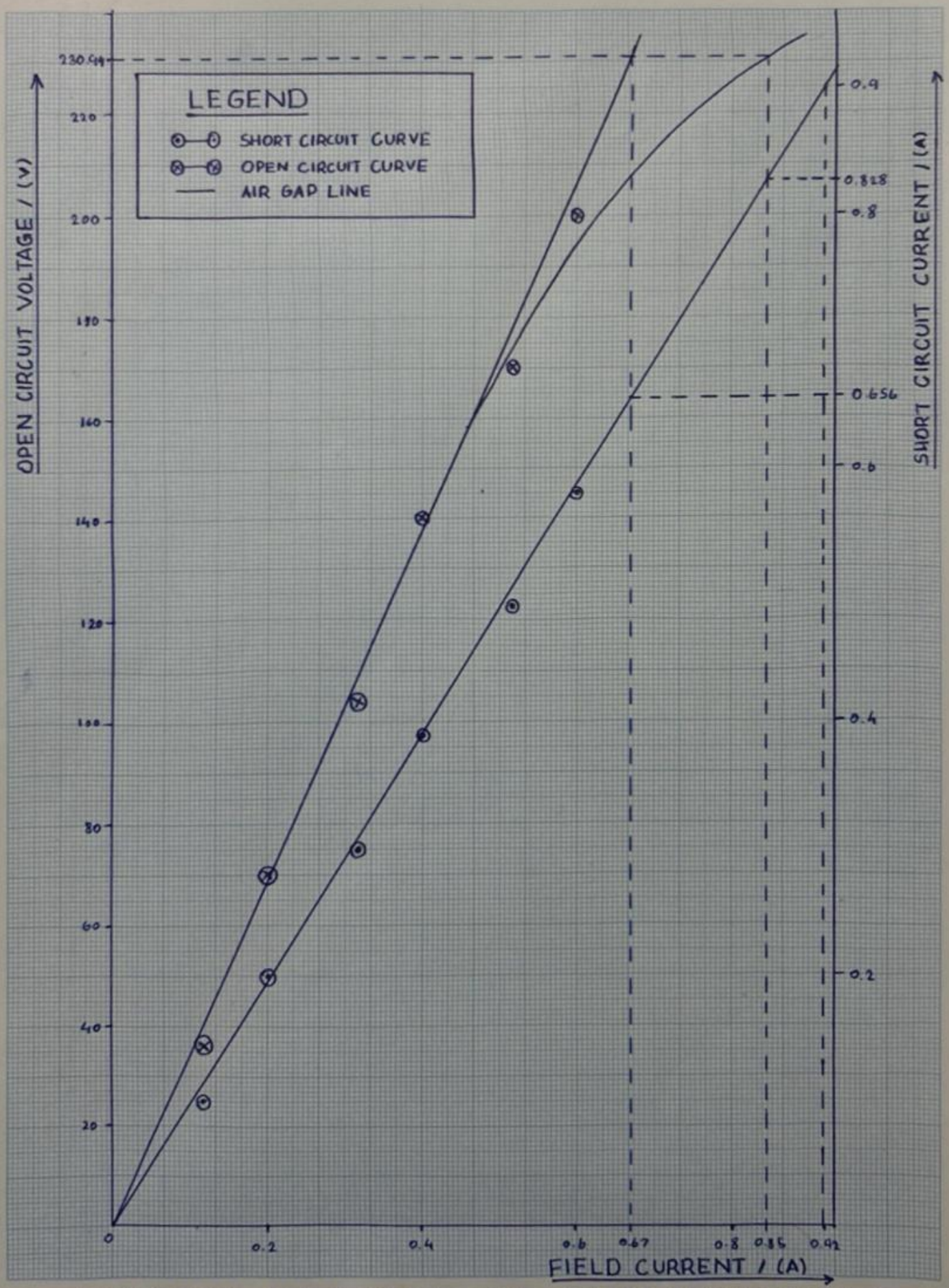


FIGURE 03 : OPEN CIRCUIT AND SHORT CIRCUIT CURVES

Scanned with CamScanner
Subject
School
Grade

OBSERVATIONS

Table 01: Open circuit test

Field Current, I_f / A	Open Circuit Voltage, V_{OC} / V
0.0	0
0.1	36
0.2	70
0.3	104
0.4	140
0.5	170
0.6	200

Table 02: Short circuit test

Field Current, I_f / A	Short Circuit Current, I_{SC} / A
0.0	0.00
0.1	0.1
0.2	0.2
0.3	0.3
0.4	0.39
0.5	0.49
0.6	0.58

CALCULATIONS

01) Assuming high X / R ratio, calculate synchronous reactance (saturated and unsaturated) using open circuit and short circuit curves.

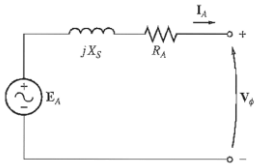


Figure 02: Equivalent Circuit of per phase Synchronous Generator

Internal impedance of the generator is,

$$Z_S = \sqrt{{R_A}^2 + {X_S}^2}$$

$$R_A \rightarrow armature\ resistance$$

$$X_S \rightarrow synchronous\ reactance$$

By assuming high X / R ratio, (neglect R_A)

$$X_S = \frac{E_A}{I_A}$$

From figure 03,

For unsaturated conditions,

$$X_{S,unsat} = \frac{E_{A,}}{I_{sc,unsat}}$$

$$X_{S,unsat} = \frac{230.94V}{0.656A}$$

$$\underline{\underline{X_{S,unsat} = 352.04\Omega}}$$

For saturated conditions,

$$X_{S,sat} = \frac{E_{A,}}{I_{sc,sat}}$$

$$X_{S,sat} = \frac{230.94\ V}{0.828\ A}$$

$$\underline{\underline{X_{S,sat} = 278.91\Omega}}$$

02) Find the value of short circuit ratio (SCR) of the machine.

$$SCR = \frac{Field\ current\ that\ produces\ rated\ voltage\ on\ open\ circuit}{Field\ current\ that\ produces\ rated\ current\ on\ short\ circuit}$$

Rated open circuit voltage = $^{400}/_{\sqrt{3}}\ V = 230.94\ V$

Rated short circuit current = $0.9\ A$

From figure 03,

$$SCR = \frac{I_{F,OC}}{I_{F,SC}}$$

$$SCR = \frac{I_{F,OC}}{I_{F,SC}}$$

$$= \frac{0.85A}{0.920A}$$

$$= \underline{\underline{0.924}}$$

03) Find the voltage regulation at full load at unity power factor and 0.8 power factor (lagging) using the synchronous impedance method.

a) <u>for unity power factor</u>

$$V_{ph} = 230.94\ V$$

$$I_a = 0.9A$$

$$X_S = 278.91\Omega$$

$$E_{ph} = \sqrt{(V_{ph}\cos\phi + I_a R_a)^2 + (V_{ph}\sin\phi + I_a X_S)^2}$$

$$= \sqrt{(230.94 + 0)^2 + (0.9 \times 278.91)^2}$$

$$E_{ph} = 341.09\ V$$

$$\text{Voltage regulation} = \frac{E_{ph} - V_{ph}}{V_{ph}} \times 100\ \%$$

$$= \frac{341.09 - 230.94}{230.94} \times 100\%$$

$$= \underline{\underline{47.70\%}}$$

b) <u>for 0.8 unity power factor lagging</u>

$$\cos\phi = 0.8, \ \ \sin\phi = 0.6$$

$$E_{ph} = \sqrt{(V_{ph}\cos\phi + I_a R_a)^2 + (V_{ph}\sin\phi + I_a X_S)^2}$$

$$= \sqrt{(230.94 \times 0.8)^2 + (230.94 \times 0.6 + 0.9 \times 278.91)^2}$$

$$= 431.17\ V$$

Voltage regulation $= \frac{E_{ph} - V_{ph}}{V_{ph}} \times 100\ \%$

$$= \frac{431.17 - 230.94}{230.94} \times 100\%$$

$$= \underline{\underline{86.73\ \%}}$$

04) Draw the performance chart of the synchronous generator.

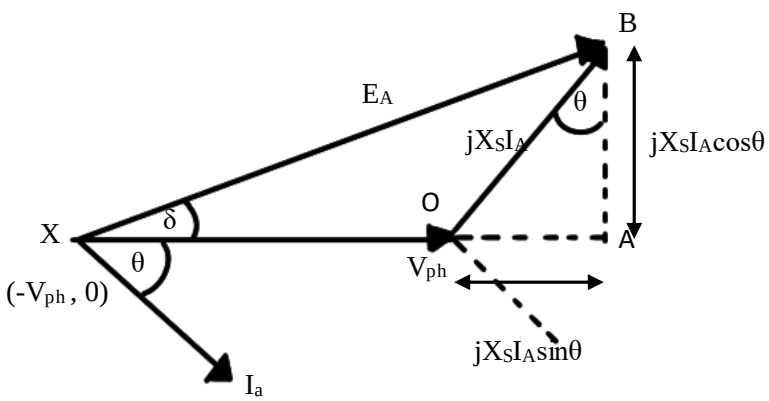
$$P = {(3V_{ph}I_a\cos\theta)}/{X_S} = S.\cos\theta$$

$$Q = {(3V_{ph}I_a\sin\theta)}/{X_S} = S.\sin\theta$$

$$S = {(3V_{ph}I_a)}/{X_S}$$

$$P_{max} = {(3V_{ph}I_a)}/{X_S}$$

Phasor diagrams in volts axes



OA = $jX_SI_A\sin\theta$

OB = $jX_SI_A\cos\theta$

$X_S >> R_A$

<u>Figure 04: phasor diagram of a synchronous generator</u>

Conversion factor to change scale of axes from voltage to power is $\frac{3V_{ph}}{X_S}$

Corresponding power diagram,

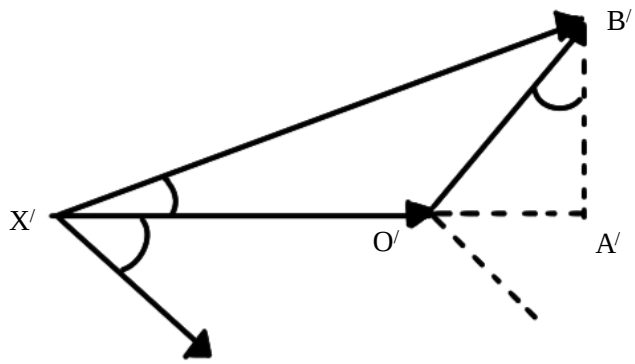


Figure 05: power diagram of a synchronous generator

$$AB^{/} = 3V_{ph} I_a \cos\theta$$

$$OA^{/} = 3V_{ph} I_a \sin\theta$$

$$X^{/}B^{/} = \frac{3V_{ph} E_a}{X_S}$$

$$O^{/}B^{/} = \frac{3V_{ph} I_a}{1} = S$$

$$X^{/} = \left(\frac{-3V_{ph}^2}{X_S}, 0\right)$$

$$X^{/} = Origin\ of\ the\ powewr\ diagram$$

$$V_{ph} = 230.94\ V$$

$$I_a = 0.9\ A$$

$$X_S = 278.91\ \Omega$$

$$\cos\phi = 0.8\ \ lagging$$

$$X^{/} = \left(\frac{-3V_{ph}^2}{X_S}, 0\right)$$

$$= \left(\frac{-3 \times 230.94^2}{278.91}, 0\right)$$

$$= (-573.51\ , 0)$$

$$O^{/}B^{/} = \frac{3V_{ph} I_a}{1} = \frac{3 \times 230.94\ V \times 0.9}{1}$$

$$O^{/}B^{/} = 0.6235 kVA$$

Rated Reactive power of the generator = S $\sin\phi$

= <u>0.3741 kvar</u>

Rated active power of the generator = S $\cos\phi$

= <u>0.4988 kW</u>

$$X^{/}B^{/} = \frac{V_{ph}E_a}{X_S}$$

$$E_a = \sqrt{(V_{ph} + I_a X_S \sin\phi)^2 + (I_a X_S \cos\phi)^2}$$

$$= \sqrt{(230.94 + 0.9 \times 278.91 \times 0.6)^2 + (0.9 \times 278.91 \times 0.8)^2}$$

$$= \underline{\underline{431.17\ \text{V}}}$$

$$D_E = \frac{3 \times 230.94}{278.91} \times 431.17$$

$$= \underline{\underline{1071.04\ \text{W}}}$$

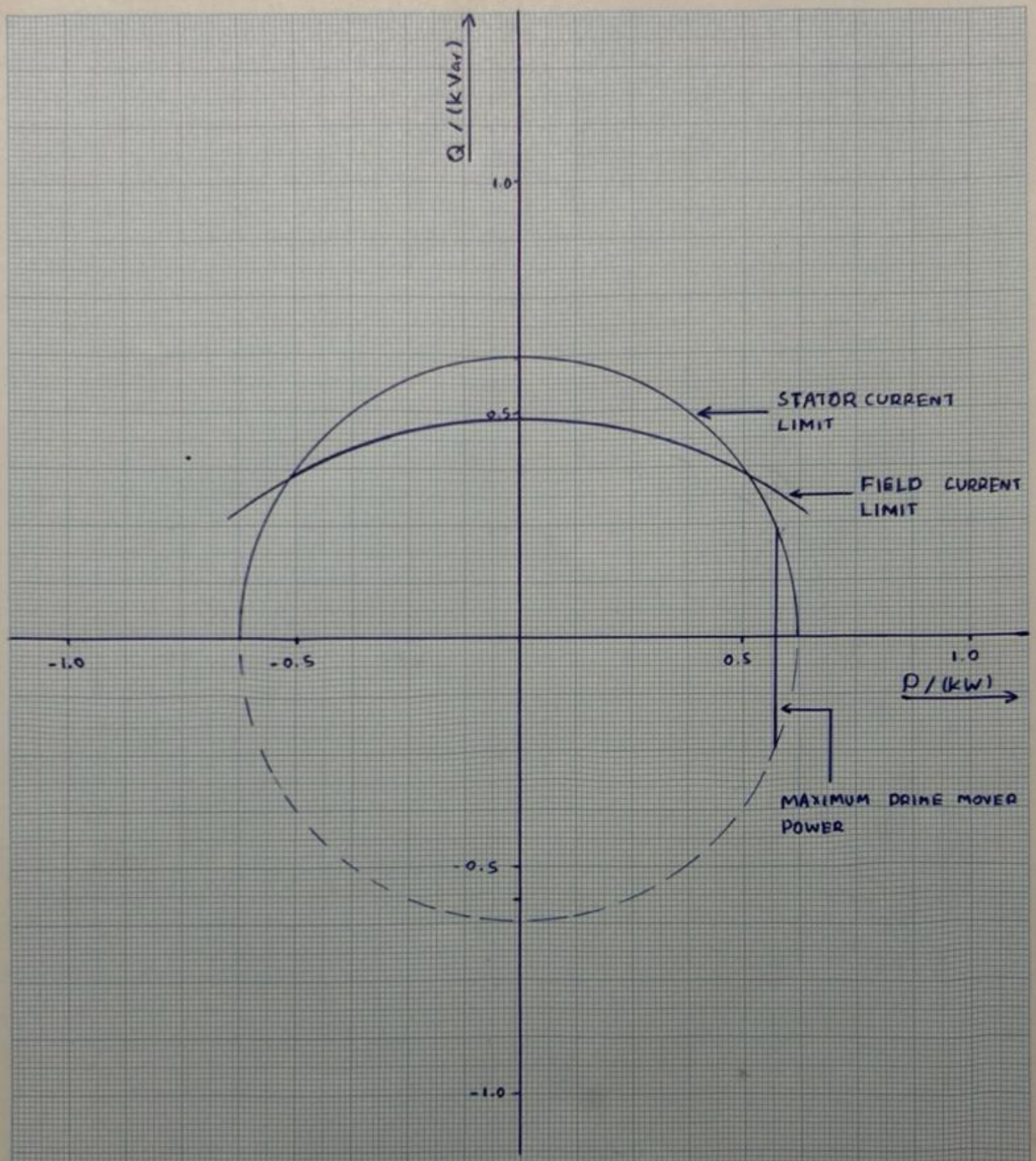


FIGURE 06: PERFORMANCE CHART OF THE SYNCRONOUS GENERATOR

1 mm Graph

Name
Subject
sarasavi BOOKSHOP
School
Grade
Scanned with CamScanner

DISCUSSION

1. What are the factors to be considered when connecting a synchronous generator to the National grid?

    - Phase Sequence
    - Magnitude of the voltage
    - Frequency
    - Phase angle

2. Mention methods to verify the phase sequences of the generator and national grid?

    Before connecting a generator to the grid, it is very important to make sure that both have the same phase sequence. If the phase sequence is wrong, it can cause mechanical stress and electrical damage. To avoid this, the phase sequence is checked using the following methods.

    1. Motor Method

    In this method, a three-phase induction motor is used.
    First, the motor is connected to the normal grid supply, and the direction of rotation of the motor is observed. This direction is taken as the reference. Then, the motor is disconnected from the grid and connected to the three-phase output of the generator. The direction of rotation is checked again. If the motor rotates in the same direction in both cases, it means the generator phase sequence is the same as the grid. If the motor rotates in the opposite direction, the phase sequence is incorrect. To correct this, any two phase terminals of the generator are interchanged, and the test is repeated until the motor rotates in the same direction as it did with the grid.

    2. Lamp method

    In the three-lamp method, three incandescent lamps are used.
    The lamps are connected across the open circuit breaker contacts as follows

    Generator Phase A to Grid Phase A
    Generator Phase B to Grid Phase B
    Generator Phase C to Grid Phase C

    The lamps act as voltage difference indicators between the generator and grid phases.
    If the phase sequence is correct the voltage difference of all three phases' changes at the same time all three lamps glow bright together and go dark together this shows that the generator and grid are in the same phase sequence. If the phase sequence is incorrect the voltage peaks occur at different times the lamps do not glow simultaneously Instead, they glow one after another in a rotating pattern this rotating glow indicates a phase sequence mismatch.

    3. Motor-Based Phase Sequence Indicator

    This method uses a phase sequence indicator meter.
    Inside the meter, there is a small three-phase motor or a light aluminum disc. The three test leads marked R, Y, and B are connected to the three-phase supply. If the phase sequence is positive (ABC), the internal disc rotates clockwise, as shown by the arrow on the meter. If the phase sequence is negative (ACB), the disc rotates counterclockwise. By checking both the grid side and generator side, we can confirm whether their phase sequences match. These methods are used to ensure that the generator and grid have the same phase sequence before synchronization. The motor method and three-lamp method are simple and easy to understand, while the phase sequence indicator provides a fast and reliable result.

3. Discuss how can you change frequency of the generated voltage?

The frequency of the voltage generated by an AC generator depends on the speed of rotation of the rotor and the number of poles of the generator. This relationship is given by

$$F_e = \frac{n_{syn}\ P}{120}$$

n_{syn} − Synchronous Speed

F_e − Electrical Frequency

P − Number of Poles

The most common and practical method is to change the speed of the prime mover driving the generator. Increasing the rotor speed increases the frequency decreasing the rotor speed decreases the frequency. In practice, this is done using a speed governor, which controls the fuel input. This method is widely used for frequency control in power systems.

From the frequency equation, it is clear that the frequency is directly proportional to the number of poles increasing the number of poles increases the frequency for a given speed decreasing the number of poles reduces the frequency. However, changing the number of poles is not practical during normal operation, as it requires a different machine design. Therefore, this method is mainly used at the design stage of the generator.

In practical systems, the frequency of the generated voltage is mainly controlled by adjusting the speed of the prime mover. The number of poles is fixed once the generator is manufactured, so it cannot be changed during operation.

REFERENCES

[1]. S. J. Chapman, *Electric Machinery Fundamentals*, 5th Ed. New York, NY, USA: McGraw-Hill, 2012.[Accessed: Jan 24, 2026]

[2]. IEEE Power System Relaying and Control (PSRC) Committee, *Practices for Generator Synchronizing Systems*, Working Group J20, IEEE, Apr. 2024. Available: https://www.pes-psrc.org/kb/report/119.pdf [Accessed: Jan. 26, 2026].

[3]. C. Joubert, "Simulation of a Pelton turbine coupled to a synchronous generator," Simulations Génie Électrique, Université Claude Bernard Lyon 1. Available: https://perso.univ-lyon1.fr/charles.joubert/web_anim/sim_Pelton_couplee_1.html [Accessed: Jan. 26, 2026].